



BUIDL CTC 2026 · DEPIN · ATTESTCOIN PROTOCOL

Ledgerline

No DePIN network can underwrite its own operators.

Ledgerline proves every earnings stream to Creditcoin and lends against the total.

No collateral. No bridge.

THE PROBLEM

Alone, every operator looks unbankable.

Someone runs a hotspot on one network, a storage node on another, a sensor on a third. Each network settles its own rewards on its own chain.

- The wireless network sees the hotspot. The storage network sees the node. Neither sees the other.
- To every individual network the operator is a stranger with nothing to pledge.
- In aggregate they are obviously good for it, and no network can see the aggregate.

HOW IT WORKS

Only a fact crosses. No token ever does.

ETHEREUM SEPOLIA

A network settles earnings and emits
`PaymentMade(operator, amount, period)`.

It never touches Creditcoin, signs nothing there, and bridges nothing.

CREDITCOIN CC3

Anyone submits an Attestcoin proof to
`IncomeRegistry.execute()`.

The `0x0FD2` precompile verifies it in that same transaction, and the settlement folds into one profile.

$$\text{limit} = \text{run-rate per period} \times (\text{history} + \text{diversification})$$

Diversification is priced, because it is real.

500 a period from one network is a worse credit than 500 across three. The first has a single point of failure: that network degrades and 100% of the income stops.

1 network	->	+0
2 networks	->	+15%
3 or more	->	+30%

No individual network can price this, because none of them can see the others. Creditcoin can, because every stream is proven into the same registry.

Three networks. One credit profile.

PROVEN REVENUE	
3,900 TST	
Wireless coverage	3,000
Decentralised storage	720
Environmental sensors	180
Run-rate	650 / period

CREDIT GRANTED	
1,007.5 TST	
Collateral posted	none
History	125%
Diversification	+30%
Drawn	250

RISK

A node that stops earning has stopped working.

There is no collateral to seize, so the risk model is cadence. For physical infrastructure that is a direct default signal rather than a proxy: revenue stops when the hardware stops.

During development the registry accepted a fully proven six-period history and **still returned a limit of zero**, because that revenue was six months stale. The freeze fired before it was tested deliberately.

Two bugs we shipped, found, and fixed.

- **Open settlement.** `pay ( )` was permissionless, so a wallet could self-transfer in a loop, netting to zero, and manufacture an earnings record for the price of gas. Now restricted to the network's paymaster. Found by running the attack, not by reading the code.
- **The credit model penalised diversifying.** Dividing by settlement count instead of distinct periods meant adding a smaller second stream lowered the limit. Only surfaced once the networks paid different amounts.

The base contract is hardened against two further flaws found in the tutorial's USCBASE and filed upstream as [gluwa/USC-Builder-Examples#37](#).

CHECK IT YOURSELF

Every claim is re-readable from public RPCs.

```
$ yarn verify
PASS Income record: 3900.0 proven across 3 networks
PASS Credit limit 1007.5 is derived from proven revenue
run-rate 650.0/period x (125% + 30%) = 1007.5
PASS Operator has drawn 250.0, nothing pledged
PASS A replayed settlement was rejected on-chain
mined and reverted in block 5429631
```

No private key, no local state. `yarn preflight` answers whether a stranger can complete the demo right now.



SCOPE, STATED PLAINLY

One source chain, three networks, a single-lender pool.

No interest, no term, no secondary market. What is submitted is the primitive:
foreign revenue, proven trustlessly, aggregated across networks, priced into
unsecured credit.

nuel-osas.github.io/ledgerline